

¹ Highlights

² **Co-firing copper in a cordierite-type glass-ceramic printed by multi-**
³ **material vat photopolymerisation: a model-based process design**

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6 Co-firing copper in a cordierite-type glass-ceramic
7 printed by multi-material vat photopolymerisation: a
8 model-based process design

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10 **Abstract**

11 *Keywords:* Co-firing, Glass-ceramic, Copper metallisation, Vat
12 photopolymerisation, Binder removal, Sintering mismatch

13 **1. Introduction**

14 Packages for millimetre-wave front ends, antenna-in-package modules and
15 wide-bandgap power devices now have to route conductors in three dimensions
16 through a low-loss dielectric, and to do it close to the die [1, 2, 3, 4]. Co-fired
17 ceramics answer part of this demand. Low-temperature co-fired ceramic
18 (LTCC) and glass-ceramic substrates combine low dielectric loss, hermeticity
19 and a thermal expansion close to that of silicon [5, 6]. They are, however,
20 built from stacked tapes: conductors are printed in planes, interconnections
21 run vertically through punched vias, and every change of geometry requires
22 new tooling [7, 5]. The Heterogeneous Integration Roadmap lists additive
23 manufacturing of ceramic packages, including lithography-based ceramic
24 manufacturing, as one way past these limits [8].

25 The conductor of choice in fired packages has long been silver, because it
26 can be co-fired in air. Silver migrates under bias and humidity and diffuses

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27 into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts
28 almost as well, costs less and is far less prone to migration, but it must be
29 fired in an atmosphere that keeps it metallic. In such an atmosphere the
30 organic binder no longer burns off cleanly: part of it is left as char, and char
31 trapped in a glass that has already sealed its pores degrades dielectric loss
32 and can bloat the part [12]. IBM solved this for its glass-ceramic/copper
33 multilayer substrates with a steam-based co-sintering process that gave a
34 dense glass-ceramic with high-conductivity copper [13, 6]. That solution was
35 developed for laminated tapes and its operating window was found empirically.

36 Vat photopolymerisation of ceramic slurries, in particular lithography-
37 based ceramic manufacturing (LCM), produces dense ceramics with features of
38 a few tens of micrometres [14, 15, 16]. It has already been used for monolithic
39 microwave filters and antenna substrates [17, 18, 19], and photocurable LTCC
40 suspensions have been formulated for it [20]. In these parts the metallisation is
41 still added afterwards, by plating or by printing on the fired ceramic [17, 21], or
42 the conductors are jetted into ceramic layers by a separate route [22, 23, 24, 25].
43 Multi-material vat photopolymerisation removes that restriction. Printers
44 such as the CeraFab Multi 2M30 switch between slurries within a layer
45 [26, 27, 28]. Co-sintered ceramic-ceramic parts have been demonstrated [29],
46 as has a zirconia part with titanium-oxide conductors co-fired in a reducing
47 atmosphere [30]. Copper slurries for vat photopolymerisation have been
48 developed separately [31, 32, 33, 34]. In principle, then, a glass-ceramic body
49 and its copper wiring can be printed as one green part. We found no study
50 of whether such a part can be fired.

51 Four requirements make that firing difficult, and they are coupled. First,

the char from both slurries must be removed before the glass seals its pores, in an atmosphere that does not oxidise the copper. Second, the two materials must shrink at compatible times and by compatible amounts; otherwise the part cambers, delaminates or cracks around its conductors [35, 36, 37]. Third, the glass must densify before it crystallises, because crystals stop viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1} between copper and a cordierite-type glass-ceramic must be accommodated on cooling. In tape-based LTCC the second requirement is met by trial, most often by adding ceramic powder to the metal paste [38] or by adjusting the heating rate [39]. Continuum models of co-sintering reproduce camber and mismatch stresses in ceramic and metal–ceramic bilayers [40, 41, 42, 43, 44, 45], but none includes the chemistry of binder removal, and none treats copper with a crystallising glass. The first and third requirements are therefore usually handled separately from the second, although in practice they compete for the same temperature range.

This paper sets out a model-based design of that firing for a copper conductor printed inside a cordierite-type glass-ceramic body. We extend a verified slab model of binder removal and sintering, originally developed for printed copper, to a crystallising glass-ceramic, and couple the two materials through the viscous analogue of laminate theory and a three-dimensional finite-element model built on the printed voxel grid. The model is verified against closed-form solutions and parameterised from published data, with every input tagged by its provenance. With it we (i) map the atmosphere and temperature window in which carbon can be removed from the glass before its pores close while copper stays metallic; (ii) show that fine copper and a

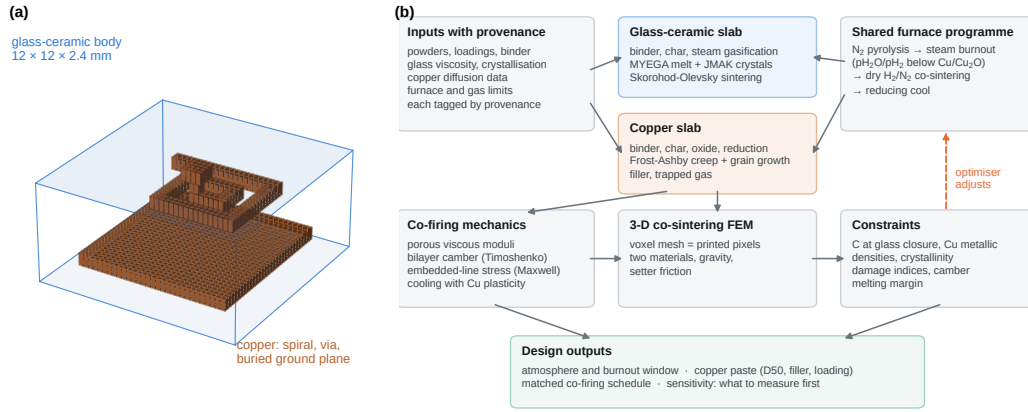


Figure 1: (a) Test vehicle: a $12 \times 12 \times 2.4$ mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

77 fragile glass densify in different temperature ranges and with different curve
 78 shapes, and quantify the consequences for camber and conductor damage;
 79 (iii) design the copper paste and the firing programme together to bring the
 80 two into step; and (iv) identify the few material parameters that decide the
 81 outcome, and should therefore be measured first.

82 2. Model formulation

83 2.1. Structure of the model

84 The co-fired part is represented at two levels (Fig. 1b). At the first level,
 85 each material is followed through the firing programme by a one-dimensional
 86 slab model that resolves heat transfer, binder decomposition, char formation
 87 and removal, pore closure and densification through the thickness of its critical

88 section. The glass-ceramic body and the copper conductor are simulated
 89 separately but share the furnace set point and the inlet gas, so both see the
 90 same thermal and chemical history. At the second level, the two densification
 91 histories are coupled mechanically: analytically, through the viscous analogue
 92 of laminate theory, and numerically, through a three-dimensional finite-element
 93 model in which both materials occupy the voxels of the printed part. Both
 94 levels use the same constitutive law for the porous body, so a free sample in
 95 the finite-element model reproduces the slab model exactly.

96 The copper slab model was developed and verified for single-material
 97 copper parts printed from a lithography-based slurry; its chemistry and its
 98 calibration against published copper data are summarised in Appendix A. The
 99 additions made here are the crystallising glass-ceramic, the copper paste with
 100 an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

101 2.2. Binder removal and residual carbon

102 The photocured binder is lumped into three pseudo-components i (a non-
 103 reactive solvent, a crosslinked network and a char-forming backbone), each
 104 decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{p,i}^2} \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{p,i}} \right) \right], \quad (1)$$

105 where $T_{p,i}$ is the temperature of the corresponding peak in a thermogravimetric
 106 trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak
 107 temperature keeps the two fitted quantities nearly independent and lets them
 108 be read directly from a single thermogravimetric curve. A fraction χ of the
 109 network and backbone mass remains as char. In a steam-bearing atmosphere

110 the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$r_g = a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}} \right)^{1/2} \frac{1 + K_{\text{H}} x_{\text{H}_2}^{\text{ref}}}{1 + K_{\text{H}} x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)} \right)_+, \quad (2)$$

111 in which c is the char concentration, $a(\varepsilon_o)$ the fraction of pore surface reachable
 112 through open porosity ε_o , f_{cl} the closed-pore fraction, K_{H} an inhibition
 113 constant for hydrogen adsorption, $Q = x_{\text{CO}}x_{\text{H}_2}/x_{\text{H}_2\text{O}}$ the reaction quotient
 114 and K_{st} the equilibrium constant from NIST-JANAF data [46]. Char in
 115 closed pores is not gasified: once the pores seal, whatever carbon remains
 116 stays in the part. The rate constant k_g is parameterised by a half-life at
 117 800 °C in the reference atmosphere and an activation energy; no measured
 118 value exists for char on either material, so both enter the sensitivity analysis.
 119 A Thiele-modulus estimate ($k_g L^2 / D_{\text{eff}} \sim 10^{-4}$ for $L = 2$ mm) shows that
 120 gasification is kinetically controlled, so steam access is represented by the
 121 open-pore accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

122 Pyrolysis gas raises the pore pressure. The overpressure is computed
 123 from the compressible Darcy equation with Kozeny–Carman and Knudsen
 124 contributions to the permeability and a polymer-permeation term that carries
 125 the flux before the pore network percolates; it is compared with the green or
 126 brown strength through a safety factor. The same slab tracks the through-
 127 thickness temperature difference and converts it into a thermal-stress index.

128 2.3. Densification

129 Both materials densify according to the Skorohod–Olevsky viscous sinter-
 130 ing model [47]. For a porous body of porosity θ and matrix shear viscosity η ,
 131 the stress is

$$\boldsymbol{\sigma} = 2\eta [\varphi(\theta) \dot{\boldsymbol{\varepsilon}}' + \psi(\theta) \text{tr}(\dot{\boldsymbol{\varepsilon}}) \mathbf{I}] + P_{\text{L}}(\theta) \mathbf{I}, \quad (3)$$

132 with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3/\theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2/r$,
 133 where γ is the surface energy and r half the Sauter mean particle diameter.
 134 A free body densifies at $\text{tr}(\dot{\epsilon}) = -(P_L - \Delta p)/(2\eta\psi)$, where Δp is the pressure
 135 of insoluble gas trapped in closed pores. The two materials differ only in η , γ
 136 and r .

137 *Glass-ceramic.* The residual melt is Newtonian. Its viscosity follows the
 138 MYEGA equation [48],

$$\log_{10} \eta_g = \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \exp \left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right], \quad (4)$$

139 with $\log_{10} \eta_\infty = -3$ (Pas), glass transition T_g (defined at 10^{12} Pas) and
 140 fragility index m . Crystallisation during firing follows non-isothermal Johnson–
 141 Mehl–Avrami–Kolmogorov kinetics [49] written in terms of the extended-
 142 volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

143 which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant
 144 $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at
 145 10 K min^{-1} peaks exactly at the measured temperature $T_{p,c}$; Kissinger’s closed
 146 form [50] is exact only for $n = 1$ and would misplace the peak by about 1 K
 147 for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m} \right)^{-[\eta]X_m} \left(1 - \frac{f_C}{0.64} \right)^{-[\eta]0.64}, \quad (6)$$

148 the Krieger–Dougherty law [51] with intrinsic viscosity $[\eta] = 2.5$ and maximum
 149 packing X_m . The second factor treats residual char particles, volume fraction
 150 f_C , the same way. Once X approaches X_m the body stops flowing, which is
 151 how crystallisation arrests densification.

152 *Copper.* The copper skeleton creeps by diffusion. Its matrix viscosity combines
 153 lattice and grain-boundary diffusion [52],

$$\eta_{\text{Cu}} = f_{\eta} \frac{k_{\text{B}} T G^2}{42 \Omega (D_{\text{v}} + \pi \delta D_{\text{b}}/G)} \left(1 + \frac{C^2}{C_{\text{inh}}^2}\right) \left(1 - \frac{f}{0.64}\right)^{-1.6}, \quad (7)$$

154 where G is the grain size, Ω the atomic volume and f_{η} a calibration factor
 155 fitted to published copper densification data (Appendix A). The carbon term
 156 C (in ppm) describes the retarding effect of residual carbon on copper, and
 157 the last factor accounts for an inert ceramic filler of volume fraction f added
 158 to the conductor paste, a standard way to slow a metal paste down to the
 159 pace of its ceramic [38]. Grain growth is parabolic, driven by grain-boundary
 160 diffusion and pinned by porosity. The filler also lowers the conductivity;
 161 together with porosity,

$$\frac{\sigma}{\sigma_0} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

162 the second factor being the Maxwell Garnett result for insulating inclusions
 163 [53].

164 2.4. Atmosphere and the metallic state of copper

165 Steam removes char but oxidises copper once the water-to-hydrogen ratio
 166 exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2 \text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp\left(-\frac{\Delta G^{\circ}(T)}{RT}\right). \quad (9)$$

167 During the burnout segment the inlet hydrogen is therefore tied to the steam
 168 content, $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_{\text{B}}))$, with a safety factor $s = 0.2$. The copper
 169 stays metallic by construction, and the steam is as oxidising towards carbon
 170 as the metal allows. This is the principle of the steam co-sintering developed

for glass-ceramic/copper substrates [13, 6]; here it is applied in a model that says how long, at what temperature and in how much steam. Carbon gasification itself is thermodynamically unconstrained over the whole window (Section 4.2); its limits are kinetic.

2.5. Mechanics of co-firing

Porous viscosities. Eq. (3) describes an isotropic, linear viscous solid with shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity $E_v = 9K_vG_v/(3K_v+G_v)$, viscous Poisson ratio $\nu_v = (3K_v-2G_v)/[2(3K_v+G_v)]$ and biaxial viscosity $B = E_v/(1 - \nu_v)$. The free linear sintering rate is $\dot{\epsilon}_f = -P_L/(6\eta\psi)$. All three quantities follow from the slab histories of T , θ , G or ξ , and carbon.

Camber. By the elastic–viscous correspondence principle, a bilayer of two linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal solution [54], with elastic moduli replaced by biaxial viscosities and thermal strain by the difference in free sintering rates [55, 37, 40]:

$$\dot{\kappa} = \frac{6(1+\lambda)^2(\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h \left[3(1+\lambda)^2 + (1+\lambda\mu) \left(\lambda^2 + \frac{1}{\lambda\mu} \right) \right]}, \quad \lambda = \frac{h_1}{h_2}, \quad \mu = \frac{B_1}{B_2}. \quad (10)$$

Layer 1 is copper and layer 2 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10) is integrated over the firing with the layer thicknesses updated by their own shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-ceramic.

Embedded conductors. A conductor line buried in the body cannot curl; its misfit is taken up by stress. We represent line and surrounding body as two

192 Maxwell elements in series, each with a porous elastic modulus and a biaxial
 193 viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

194 A compliant line follows the body and carries B_{Cu} times the misfit rate; a
 195 stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$)
 196 opens voids and breaks the line; compression in the copper means the body is
 197 shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin
 198 of radial cracks around conductors [35, 36]. Two damage indices compare
 199 these stresses with the strength of the weaker partner at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

200 where the strengths σ_s grow from the binder-carried green strength through
 201 the neck-controlled brown strength. The series coupling carries an Eshelby
 202 factor of order unity [56] that we do not resolve, so the indices are used
 203 comparatively. The same coupling is resolved exactly in three dimensions
 204 below.

205 *Three-dimensional co-sintering.* The printed part is meshed voxel by voxel
 206 with trilinear hexahedra, so the mesh is the printed pixel grid. Each element
 207 carries its material, porosity and microstructural state. The velocity field sat-
 208 isfies $\nabla \cdot \boldsymbol{\sigma} + \rho_b \mathbf{g} = \mathbf{0}$ with Eq. (3), gravity, and a setter plate beneath the part:
 209 zero normal velocity and regularised Coulomb friction ($\mu = 0.6$) iterated to
 210 convergence at each step. Volumetric terms use one-point quadrature, which
 211 avoids locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [57] and
 212 solved by conjugate gradients preconditioned with smoothed-aggregation alge-
 213 braic multigrid carrying the six rigid-body modes [58]. Porosity is advanced

214 with the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\epsilon} \Delta t)$. The
 215 temperature and carbon histories are taken from the two slab simulations.

216 *Cooling.* Below a stress-free temperature T_{sf} , where the residual glass stops
 217 relaxing, the thermal-expansion mismatch is carried elastically by the glass-
 218 ceramic and elastoplastically by the copper. The copper layer is treated as
 219 elastic-perfectly plastic with a temperature-dependent yield stress; its mean
 220 stress is capped at yield by a uniform plastic strain, and the curvature follows
 221 from Timoshenko’s elastic solution for the remaining elastic misfit.

222 2.6. Numerical solution, design search and sensitivity

223 The slab equations are integrated with a linearly implicit Rosenbrock
 224 method of orders 2(3) [59] on a geometrically graded mesh that is finest at
 225 the surface. A co-firing evaluation, two slab simulations and the mechanical
 226 post-processing, takes about 6 s. Designs were searched by differential evolu-
 227 tion [60] over eleven variables: copper median particle size, filler fraction and
 228 solids loading; burnout temperature, time and steam fraction; temperature
 229 and duration of an intermediate matching hold; heating rate to the peak; peak
 230 temperature and hold. The objective was the largest free-strain mismatch be-
 231 tween the two materials during firing, $\max_t |\varepsilon_{\text{f,GC}} - \varepsilon_{\text{f,Cu}}|$, which governs both
 232 camber and conductor stress. It was minimised subject to hard constraints
 233 and to a cycle-time budget of 18 h, the ε -constraint form of the trade-off
 234 between cycle time and mismatch. The constraints were: carbon at glass pore
 235 closure below 100 ppm and in the copper below 200 ppm; relative densities
 236 of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised
 237 fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure

above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. The population of 88 designs was evolved for 15 generations.

The influence of uncertain inputs was screened with Morris elementary effects [61]: 16 trajectories through a four-level grid over fourteen factors, 240 co-firing evaluations in total, around the optimised design. Twelve factors are the uncertain inputs with the ranges of Table 1; the other two are manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size and ± 0.05 in filler fraction. For each output the mean absolute elementary effect μ^* measures the overall influence of a factor and the standard deviation σ its non-linearity and interaction with other factors.

3. Parameters and calibration

3.1. Provenance of the inputs

No measurements were made for this study. Every input therefore comes from one of five sources, and Table 1 states which one applies. A value is labelled L when it was reported for the glass-ceramic of the IBM process or for copper itself; A when it was taken from an analogous material, usually a commercial copper-compatible LTCC; C when it was calibrated here against published process observations (Section 3.2); D when it follows from other entries of the table; and E when it is an engineering estimate. Estimated and analogue values that matter enter the sensitivity analysis of Section 4.10, with the ranges given in the table.

The reference glass is the MgO-rich cordierite-type glass used by IBM

for its copper-compatible substrates (glass #12: 52.5 SiO₂, 22.0 MgO, 22.0 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [62, 63]. It was chosen because it is the only copper-compatible glass-ceramic for which the firing process, the atmosphere and the reasons behind the chosen temperatures have been published in detail. Its viscosity has not been published in a form we could read, so it was calibrated.

3.2. Calibration of the glass against the IBM process

The patents that describe the IBM steam co-sintering process state, for glass #12, why the burnout hold sits where it does. The hold is placed at 785 ± 10 °C, between the annealing and softening points of the glass. At 750 °C carbon removal would take “prohibitively excessive amounts of time”; at 830 °C the glass pores close and trap water and binder residue [62]. A later patent from the same group reports that 33–50% of the total fired shrinkage occurs during the burnout step [78], and a third places the onset of crystallisation at about 900 °C, after the burnout hold and before the crystallisation hold at about 960 °C [79]. We simulated the full worked schedule of Ref. [62] (N₂ to 200 °C, H₂/H₂O to 450 °C, 2.9 K min⁻¹ to the hold, 6 h at H₂/H₂O = 10⁻⁴, 0.5 h in N₂, 2.1 K min⁻¹ to 960 °C, 2 h) and used these statements as targets. The patent specifies the H₂/H₂O ratio but not the dilution; we assumed 50% steam in nitrogen.

Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C produce 40% of the fired shrinkage with the pores still open, and that the pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_{\infty} = -3$, has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used

287 in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity
 288 falls by one decade per 43 K. Giess et al. found that the sintering of an IBM
 289 cordierite-type glass over the same range followed the viscosity of the glass,
 290 which fell by about one decade per 40 K [65]. The annealing point, taken as
 291 10^{12} Pa s, lies at 734 °C and the dilatometric softening point ($10^{6.6}$ Pa s) at
 292 980 °C, which brackets the 785 °C hold as the patent requires. Stoichiometric
 293 cordierite glass is one to four decades more viscous over 780–960 °C [80].
 294 Glass #12 contains more MgO and less Al_2O_3 than cordierite, together with
 295 B_2O_3 and P_2O_5 , so a lower viscosity is plausible, but this comparison is not a
 296 test of the fit.

297 The crystallisation kinetics were treated in the same way. With the
 298 activation energy measured on an IBM-type glass ($E_c = 303 \text{ kJ mol}^{-1}$ [67])
 299 and $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in
 300 the worked schedule requires an exotherm peak at 1040 °C at 10 K min^{-1} .
 301 The simulated exotherm then spans 978–1066 °C, a width of about 90 K,
 302 which agrees with the 80–100 °C window that the IBM patent associates with
 303 the crystallisation exotherm of its glasses [63]. Differential thermal analysis of
 304 other MgO– Al_2O_3 – SiO_2 glasses places the α -cordierite peak at 1006–1032 °C
 305 [64, 69].

306 Table 2 summarises the result. The 785 °C hold removes the char and
 307 leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported
 308 for the process, while the pores stay open. At 830 °C the body densifies
 309 completely during the hold and seals its pores. These two rows reproduce the
 310 calibration targets and are therefore not independent evidence. The 750 °C
 311 row is: it depends on the gasification kinetics, which were not fitted, and

312 it leaves four orders of magnitude more carbon than the 785 °C hold. The
 313 model does not reproduce the absolute time scale of the patent’s statement.
 314 With the estimated gasification rate, 33 ppm of carbon remain after 6 h at
 315 750 °C, which may or may not have been acceptable for IBM; no carbon
 316 specification was published. For all three holds the final body is 99% dense
 317 and fully crystallised, and its free linear shrinkage is 23.1%.

318 4. Results

319 4.1. Verification

320 The model was checked against closed-form solutions before it was used
 321 for design (Appendix B). Isothermal densification of the glass-ceramic follows
 322 the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16
 323 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-
 324 ceramic plate converges to the viscous Timoshenko value at second order,
 325 with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical camber of
 326 Eq. (10) and the finite-element model therefore describe the same mechanics,
 327 and the analytical form can be used inside the design search.

328 4.2. Atmosphere window for burnout

329 Fig. 3 places the burnout atmosphere in the plane of temperature and
 330 $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Copper remains metallic below the orange line, which falls from
 331 $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by
 332 steam is thermodynamically possible above the dashed line, which lies eight
 333 or more decades lower. The two constraints therefore leave a wide band of
 334 permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30%

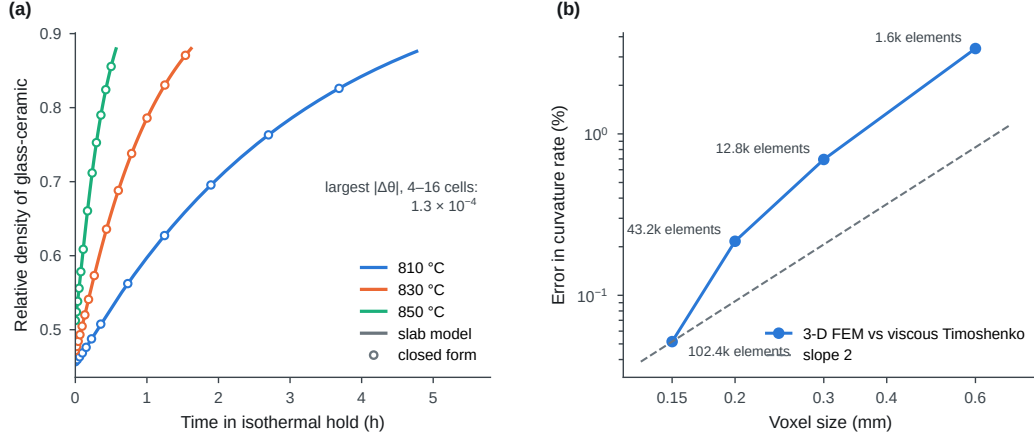


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12×12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

335 steam with hydrogen at 20% of the copper limit ($p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$) gives
 336 a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler
 337 at 20 °C, a common choice for copper, contains 2.3% water; its half-life at
 338 the same temperature is 3.1 h, seven times longer, because the steam partial
 339 pressure is low and hydrogen inhibits the reaction.

340 The operating point of the IBM process, $\text{H}_2/\text{H}_2\text{O} = 10^{-4}$ at 785 ± 10 °C
 341 [62], lies at 17% of the copper limit and almost coincides with the steam
 342 point in Fig. 3. The rule used here, hydrogen tied to steam so that the
 343 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, therefore
 344 reproduces the atmosphere that was found empirically for tape-cast substrates.
 345 It also says how the ratio should change if the burnout temperature changes:

346 by a factor of about 2.5 per 100 K, following $K_{\text{Cu}}(T)$.

347 4.3. Removing carbon before the glass seals

348 The char has to leave the glass before its pores close, and the glass
349 carries far more of it than the copper does. After pyrolysis in nitrogen the
350 glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper
351 5 300 ppm. The difference follows from the densities: both slurries hold a
352 similar volume of binder, but a unit volume of glass-ceramic solids weighs
353 a third as much as the same volume of copper. Carbon on the glass is also
354 the more harmful, since whatever is sealed in reacts with the water released
355 during crystallisation and bloats the part [62].

356 Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores
357 close, as a function of burnout temperature and hold time. In steam (Fig. 4a)
358 the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the
359 window closes abruptly at 850 °C, where the glass seals during the hold. The
360 IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In
361 humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings
362 the carbon down to 350 ppm; every other combination leaves more than
363 1000 ppm. Residual carbon also slows the densification of the glass. Acting
364 as rigid inclusions, the 0.3–2% of carbon retained in forming gas lowers the
365 final density from 0.99 to 0.96–0.99. Steam is therefore not an option for this
366 material pair but a requirement.

367 4.4. Two densification ranges

368 Fig. 5 compares the free sintering of the two materials after binder removal
369 at 5 K min⁻¹ in dry hydrogen. The glass-ceramic shrinks by 23% between 850

370 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises:
371 93% of the glass has crystallised at 1060 °C. Fine copper densifies much
372 earlier, and over a much wider range. A 3 μm powder shrinks between 454
373 and 702 °C; a 1 μm powder between 323 and 463 °C. Only for powders of
374 12 μm or coarser does the midpoint of densification fall inside the range of
375 the glass, and these powders do not reach high density before 1060 °C: 83%
376 for 12 μm and 60% for 20 μm .

377 The two curves also differ in shape. The glass densifies over a range of
378 about 80 K, as a Newtonian melt whose viscosity falls steeply with tempera-
379 ture. Copper densifies over 140–350 K, because diffusional creep has a lower
380 activation energy and grain growth slows it as it proceeds. Adding an inert
381 filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler
382 shifts the midpoint of 3 μm copper from 573 to 657 °C and widens the 5–95%
383 range from 248 to 367 K. Neither particle size nor filler, on its own, gives
384 copper a shrinkage curve that is as late and as steep as that of the glass. The
385 matching has to come partly from the furnace programme.

386 4.5. *Co-firing with an unmatched paste*

387 The baseline design combines a 3 μm copper paste without filler with a
388 programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a
389 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4%
390 H_2 (Fig. 6a). Chemically, the design works. The glass-ceramic retains 33 ppm
391 of carbon when its pores close and reaches 99.1% density and full crystallinity;
392 the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96%
393 IACS.

394 Mechanically, it fails. As soon as steam begins to remove the char, the

395 copper starts to sinter, while the glass is still rigid. At the end of the burnout
 396 hold the logarithmic shrinkage of the copper is 21% and that of the glass 7%;
 397 the free-strain mismatch peaks at 15.6%, and a free bilayer of 0.1 mm copper
 398 on 1 mm glass-ceramic would reach a curvature of 58 m^{-1} , concave towards
 399 the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at
 400 680°C , during the ramp to the burnout temperature, when the line is pulled
 401 in tension by the rigid body around it while its own necks are still weak. The
 402 glass then catches up. Its shrinkage overtakes that of the copper at 889°C ,
 403 the bilayer bends the other way and reaches -259 m^{-1} (a radius of 3.9 mm)
 404 at 930°C , and the final shrinkages differ by 4%: 26.3% for the glass-ceramic
 405 against 22.3% for the copper, both as logarithmic strains. For this material
 406 pair the difficulty lies less in the final shrinkage than in the order in which
 407 the two materials shrink.

408 *4.6. Copper paste design maps*

409 *4.7. A matched paste and programme*

410 *4.8. The printed test vehicle*

411 *4.9. Cooling from the stress-free temperature*

412 Below about 690°C , the strain point of the calibrated glass, mismatch
 413 stresses no longer relax, and the expansion mismatch between copper (16.5–
 414 22 ppm K^{-1} over this range) and the glass-ceramic (3 ppm K^{-1}) is carried
 415 elastically. Fig. 7 shows the consequences for a copper plane on 1 mm of glass-
 416 ceramic. If copper remained elastic, its mean stress would exceed 900 MPa
 417 at room temperature and a 20 mm stack with $100 \mu\text{m}$ of copper would bow
 418 by $250 \mu\text{m}$. Annealed copper cannot carry that stress. It yields within

the first 10 K of cooling and then follows its yield stress, which rises from about 25 MPa at 690 °C to 69 MPa at room temperature. The bow falls accordingly, to 13 μm for 100 μm of copper and to 6 μm for 50 μm , and the glass-ceramic stays in mild compression (-7 MPa for 100 μm of copper). Plastic accommodation makes the thermal-expansion mismatch a secondary problem compared with the sintering mismatch, provided the copper is dense and ductile. The price is cyclic plasticity in the conductor, which matters for thermal cycling in service but is outside the scope of this study.

4.10. Which inputs decide the outcome

5. Discussion

6. Conclusions

CRediT authorship contribution statement

[Author names]: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, input parameters with their provenance, and the scripts that generate every figure in this paper are available from the corresponding author on reasonable request.

441 **Declaration of generative AI and AI-assisted technologies in the**
442 **writing process**

443 During the preparation of this work the authors used [name of tool] in
444 order to [reason, e.g. assist with code development, data analysis and drafting
445 of the text]. After using this tool, the authors reviewed and edited the content
446 as needed and take full responsibility for the content of the publication.

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Appendix A. Copper slab model and its calibration

The copper conductor uses the slab model developed for single-material copper parts printed from a photocurable slurry. Besides the binder chemistry of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None of these oxidation paths is active in the co-firing programmes studied here, because the inlet gas contains

no oxygen and the water-to-hydrogen ratio is held below the copper–oxide equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses the lattice and grain-boundary diffusion data for copper tabulated by Frost and Ashby [52], with one factor, f_η , fitted to the densification of copper parts printed from a 60 vol% photocurable slurry of 22 μm powder and sintered for 4 h at 1050 $^\circ\text{C}$ in hydrogen, which reached 90–94% relative density [31]. The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range. The powder size enters Eq. (7) through the grain size, set initially to 0.4 times the median particle size, and through the sintering stress. Transferring f_η from 22 μm to the 3–12 μm powders considered here therefore rests on the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity analysis of Section 4.10.

Appendix B. Verification

Viscous densification of the glass-ceramic. With crystallisation and char removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} = -9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830 and 850 $^\circ\text{C}$ with 4, 8 and 16 cells. The simulated porosity follows the closed form, integrated along the simulated part temperature, to within 1.7×10^{-4} over the full range of densification before pore closure, independently of the mesh (Fig. 2a).

Bilayer curvature. The three-dimensional model of a free 12 $\times$ 12 mm plate made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instantaneous velocity field at 900 $^\circ\text{C}$, and the curvature rate was fitted to the mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10)

777 with second-order behaviour: the error falls from 3.4% at 0.6 mm voxels to
778 0.69, 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

779 *Free sintering and mass conservation.* A free, frictionless cube held isother-
780 mally densifies homogeneously and isotropically, matches the closed-form
781 Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to
782 0.07%.

783 *Crystallisation kinetics.* With the rate constant calibrated as described in
784 Section 2.3, the simulated exotherm at 10 K min^{-1} peaks within 0.01 K of
785 the prescribed temperature for Avrami exponents between 1 and 4.

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

Quantity	Symbol	Value	Basis	Source, remark or screening
<i>Glass-ceramic body</i>				
Median particle size	d_{50}	3.0 μm	A	IBM glass powders 2–7 μm
Size span $(d_{90} - d_{10})/d_{50}$	–	1.5	E	
Solids loading of the slurry	ϕ_{GC}	0.45	E	0.40–0.50
Density of the dense body	ρ_{s}	2600 kg m^{-3}	A	dense α -cordierite glass-cer [64]
Surface energy	γ	0.36 J m^{-2}	L	calculated for a cordierite-ty 0.36, since water vapour low sion of silicate glass [66]
Glass transition (10^{12} Pa s)	T_{g}	734 $^{\circ}\text{C}$	C	714–754 $^{\circ}\text{C}$
Fragility index	m	32.4	C	28–40
High-temperature viscosity limit	$\log_{10} \eta_{\infty}$	–3	L	[48]
Crystallisation activation energy	E_{c}	303 kJ mol^{-1}	L	IBM-type high-cordierite 440 kJ mol^{-1} in other MgO– [68, 69]; 250–470 kJ mol^{-1}
Avrami exponent	n	2	E	1.5–3.7; 2.8–3.7 in a related glass [69]
Exotherm peak at 10 K min^{-1}	$T_{\text{p,c}}$	1040 $^{\circ}\text{C}$	C	1000–1080 $^{\circ}\text{C}$
Maximum crystal packing	X_{m}	0.64	E	random close packing
Density at pore closure	ρ_{cl}	0.92	E	
Thermal conductivity, dense	k	2.8 $\text{W m}^{-1} \text{K}^{-1}$	A	copper co-fired LTCC GL57
Young’s modulus, dense	E_{GC}	128 GPa	A	GL570 [70]
Poisson’s ratio	ν_{GC}	0.25	E	
Thermal expansion coefficient	α_{GC}	3.0 ppm K^{-1}	L	production glass-ceramic su
Flexural strength	$\sigma_{\text{s,GC}}$	210 MPa	L	glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [
Stress-free temperature	T_{sf}	690 $^{\circ}\text{C}$	D	strain point ($10^{13.5}$ Pa s) of
<i>Copper conductor</i>				
	37			
Median particle size	d_{50}	1–20 μm	–	design variable; baseline 3 μm
Size span	–	1.2	E	
Solids loading	ϕ_{Cu}	0.40–0.60	–	design variable; baseline 0.5
Inert filler fraction	f	0–0.40	–	design variable; baseline 0
Viscosity factor	f_{η}	0.28	C	Appendix A, from [31]; 0.1

Table 2: The IBM worked schedule [62] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 °C	785 °C	830 °C
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [62]	too slow	chosen	pores close

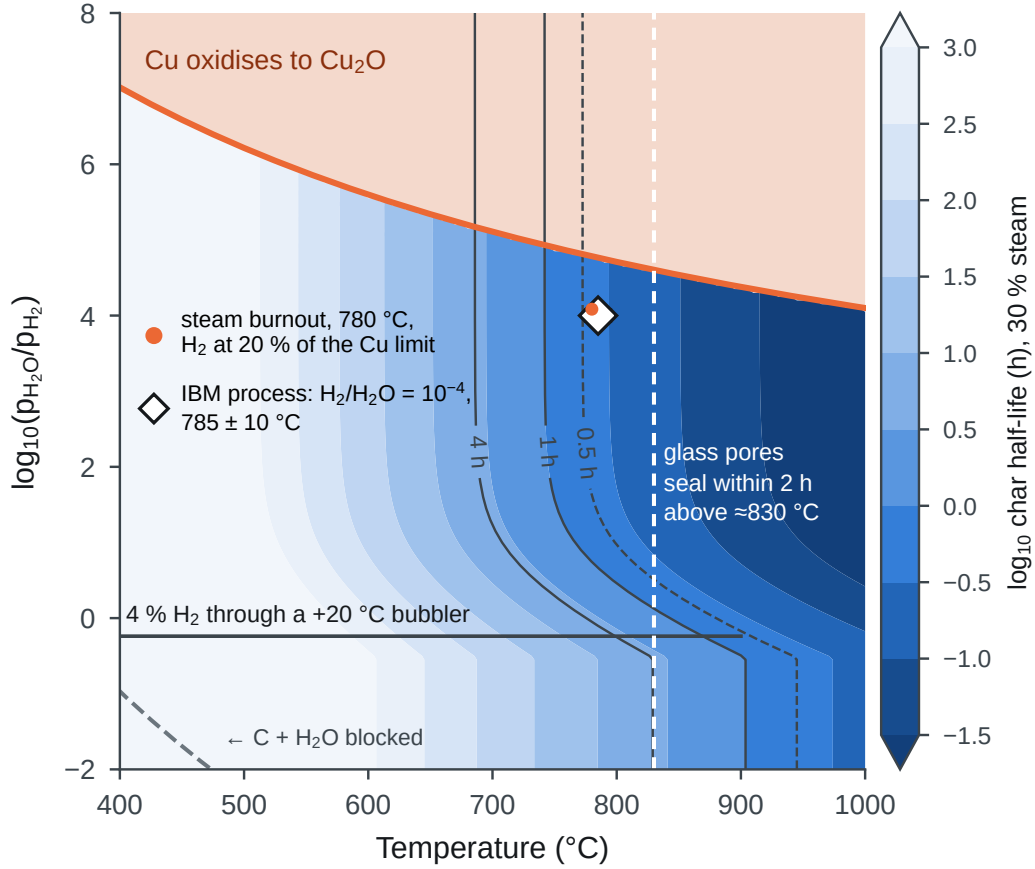


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{\text{CO}} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [62]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

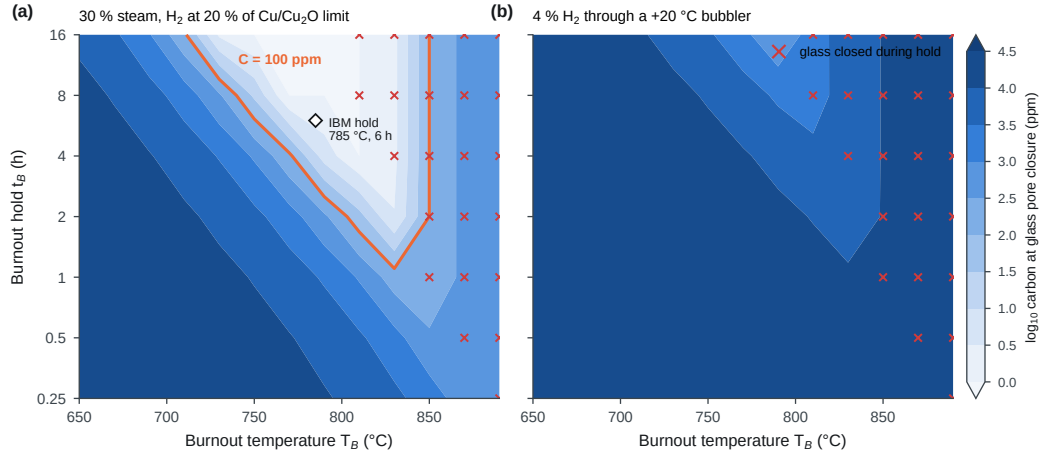


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu_2O limit; (b) 4% H_2 humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [62].

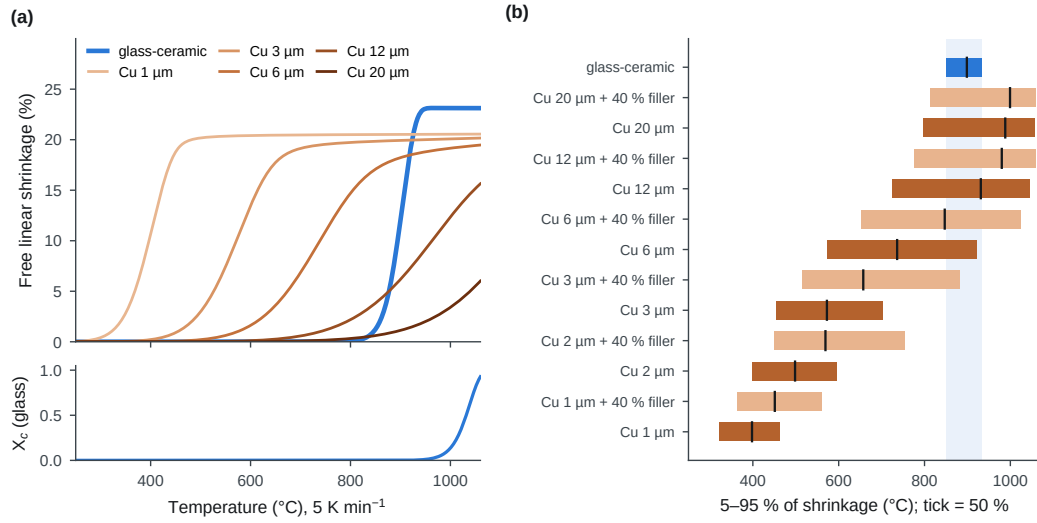


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

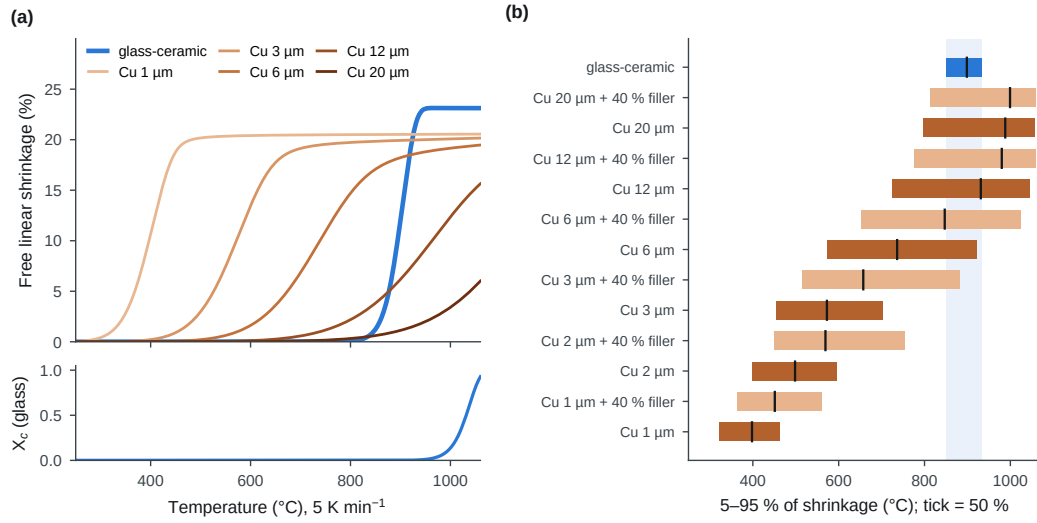


Figure 6: Co-firing histories. (a) Baseline: 3 μm copper without filler, 4 h steam burnout at 780 °C, 1 h at 960 °C. (b) Optimised paste and programme (Table ??). From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N_2 ; green, steam; blue, dry H_2/N_2); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

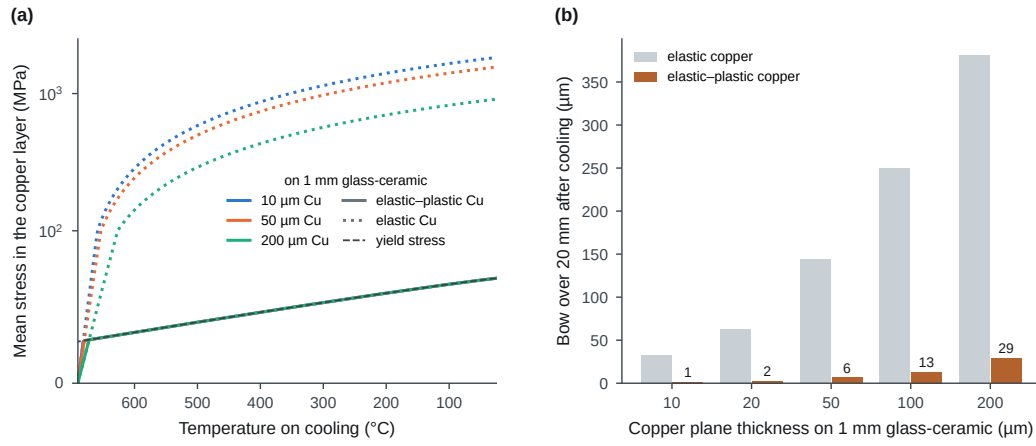


Figure 7: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690°C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.